

Symbolic Mathematics

Mathematics & Artificial Intelligence:-

Mathematics and AI are supposed to be the two branches of same tree which are interlinked with each other to master them in both of them.

Mathematics help you to design the basic rules of AI.

On the top of those mathematics rule the algorithm of AI are designed. So, to master AI you must need to master the mother of all sciences called Mathematics.

Symbolic Mathematics:-

In mathematics and computer science it is a scientific area that refers to the study and development of algorithms and software for manipulating mathematical expressions and other objects.

Symbolic calculations use symbols.

Symbolic Computation:-

It is the sub-area of mathematics and computer science which solves

problems on symbolic objects representable on computer.

Typical examples of some objects are:

→ Algebraic Expressions

→ Logical Proposition and program themselves.

Translating English Equations:-

Some operators are use to translate English sentences or equations into symbolic form.

$\sim$	tilde	Negation
$\cdot$	dot	Conjunction
$\vee$	wedge	Disjunction
$\supset$	horseshoe	Implication
$\equiv$	triple bar	Equivalence

Capital letters stand for simple statements. Simple statements are statements that cannot be broken down any further and still be truth functional.

Examples:-

- I am either clogging or singing in the raining. ($C \vee S$)
- I will jump off the clip if and only if you do ($I \equiv Y$)

First Order Predicate Logic:-

Propositional logic assumes the world contains facts that are true or false.

It is first order logic.

Object:- People, houses, numbers, color, baseball.

Relations between objects:- red, round, prime, brother of, bigger than, part of, comes, between.

Example:-

Gorilla is black

$Gorilla(x) \rightarrow black(x)$

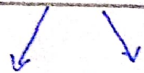
All boys like cricket.

$\forall x: boys(x) \rightarrow like(x, cricket)$

Some boys like football.

$\exists x: boys(x) \wedge like(boys, football)$

Quantifiers



Existential
quantifier

Universal
quantifier

Translating words to Algebraic Expressions:-

Five more than three times a number is twenty six. What is the value of the number?

The equation would be $3x+5=26$.
Find the value of x .

$$3x+5=26$$

$$3x=26-5$$

$$3x=21$$

$$x=21/3$$

$$x=7.$$

Solving algebraic equations symbolically:-

We know that, algebraic equations can be solved by three methods:

→ Graphically

→ Substitution

→ Elimination

Substitution Method:-

$$3x+2y=2 \quad \text{--- (i)}$$

$$y=2x-6 \quad \text{--- (ii)}$$

Substituting value of y in (i)

$$3x+2(2x-6)=2$$

$$3x+4x-12=2$$

$$7x=2+12$$

$$x=14/7$$

$$x=2$$

$$y=2x-6$$

$$y=2(2)-6$$

$$y=-2$$

Elimination Method:-

$$3x - y = 5$$

$$x + y = 3$$

$$4x = 8$$

$$x = 2$$

$$3(x) - y = 5$$

$$3(2) - y = 5$$

$$-y = 5 - 6$$

$$-y = -1$$

$$y = 1$$